

Interpolation with trigonometric polynomials

f periodic function

$$f(0) = f(2\pi)$$

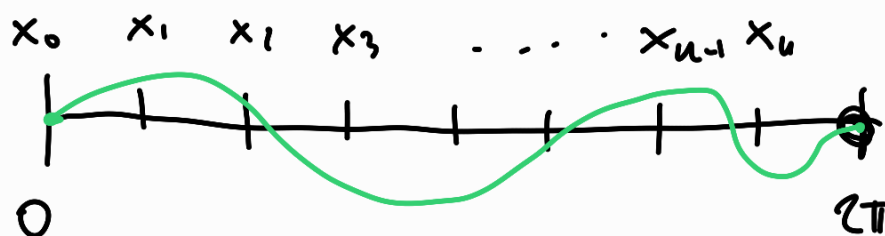
$$f(0) = f(2\pi \cdot n) \quad n \in \mathbb{Z}$$

$\tilde{f}(x)$ trigonometric interpolator

$$\tilde{f}(x) = \frac{a_0}{2} + \sum_{k=1}^n [a_k \cos(kx) + b_k \sin(kx)]$$

$$\{x_j\}_{j=0}^n \subseteq [0, 2\pi] \quad x_j = \frac{2\pi j}{n+1} \quad j = 0 \dots n$$

$$h = x_{j+1} - x_j \quad j = 0 \dots n$$



$$\tilde{f}(x_j) = f(x_j) \quad \forall j = 0 \dots n$$

Interpolation constraint

$\tilde{f}$ which is a trigonometric polynomial
and interpolate f at the $n+1$ points x_j

Euler's formula $e^{ikx} = \cos(kx) + i \sin(kx)$
 $u = 2M$ for $M \in \mathbb{N}$

$$\tilde{f}(x) = \sum_{k=-M}^M c_k e^{ikx}$$

$$\tilde{f}(x) = \frac{a_0}{2} + \sum_{n=1}^u \left[a_n \cos(nx) + b_n \sin(nx) \right]$$

$$\begin{cases} a_n = c_n + c_{-n} \\ b_n = i(c_n - c_{-n}) \end{cases} \quad \begin{cases} c_n = \frac{1}{2}(a_n - i b_n) \\ c_{-n} = \frac{1}{2}(a_n + i b_n) \end{cases}$$

Computing the coefficients of $\tilde{f}$

$$\sum_{k=-M}^M c_k e^{ikx_j} = \tilde{f}(x_j) = f(x_j) \quad \forall j = 0 \dots u$$

$u+1$ equations

$u+1$ unknowns c_k

$$c_m = \frac{1}{u+1} \sum_{j=0}^u f(x_j) e^{-imx_j}$$

$$\underline{c} = \begin{bmatrix} c_{-M} \\ \vdots \\ c_M \end{bmatrix} \quad \underline{f} = \begin{bmatrix} f(x_0) \\ \vdots \\ f(x_u) \end{bmatrix}$$

$$\underline{C} = T \underline{f} \quad T_{kj} = \frac{1}{n+1} e^{-ikj\Delta}$$

Discrete Fourier Transform
DFT

$$\underline{f} = T^{-1} \underline{C} \quad (T^{-1})_{kj} = e^{ikj\Delta}$$

Inverse DFT
IDFT

complexity of DFT / IDFT

$$\underline{y} = A \underline{x} \Leftrightarrow y_i = \sum_{j=1}^n A_{ij} x_j \quad i=1 \dots n$$

$\sim n^2$ flop

better than $\frac{n^3}{3}$ for a general linear system

$$T = \frac{1}{n+1} \begin{bmatrix} e^{-i\Delta} & e^{-2i\Delta} & e^{-3i\Delta} & e^{-4i\Delta} & \dots \\ e^{-2i\Delta} & e^{-4i\Delta} & e^{-6i\Delta} & e^{-8i\Delta} & \\ e^{-3i\Delta} & e^{-6i\Delta} & e^{-9i\Delta} & e^{-12i\Delta} & \dots \\ e^{-4i\Delta} & & & & \end{bmatrix}$$

T matrix with a special structure

computing $T \underline{f}$ with T Toeplitz
has a cost $n \log_2 n$ flops

Exploiting the Toeplitz form of T
to compute DFT and IDFT
Is called

FFT Fast Fourier Transform

Cooley and Tukey in 1965

Express the trigonometric interpolator in Lagrangian form

$$\tilde{f}(x) = \sum_{k=-M}^{+M} c_k e^{ikx} \quad (1)$$

using DFT $c_k = \frac{1}{n+1} \sum_{j=0}^n f(x_j) e^{-ik_j h} \quad (2)$

substituting (2) in (1)

$$\tilde{f}(x) = \sum_{j=0}^n \underline{f(x_j)} \left(\underbrace{\frac{1}{n+1} \sum_{k=-M}^M e^{ik(x-jh)}}_{\varphi_j^T(x)} \right)$$

$\varphi_j^T(x)$ Lagrangian basis

$$\varphi_j^T(x_n) = \delta_{jk} = \begin{cases} 1 & \text{if } j=k \\ 0 & \text{otherwise} \end{cases}$$

Accuracy of trigonometric interpolation

Let $f(x)$ be periodic with period $[0, 2\pi]$

Let $f(x) \in C^s([0, 2\pi])$

Let $f^{(i)}(x)$ be periodic for $i=1, \dots, s$

$$\max_{x \in [0, 2\pi]} |f(x) - \tilde{f}(x)| \leq C \frac{1}{n^s}$$

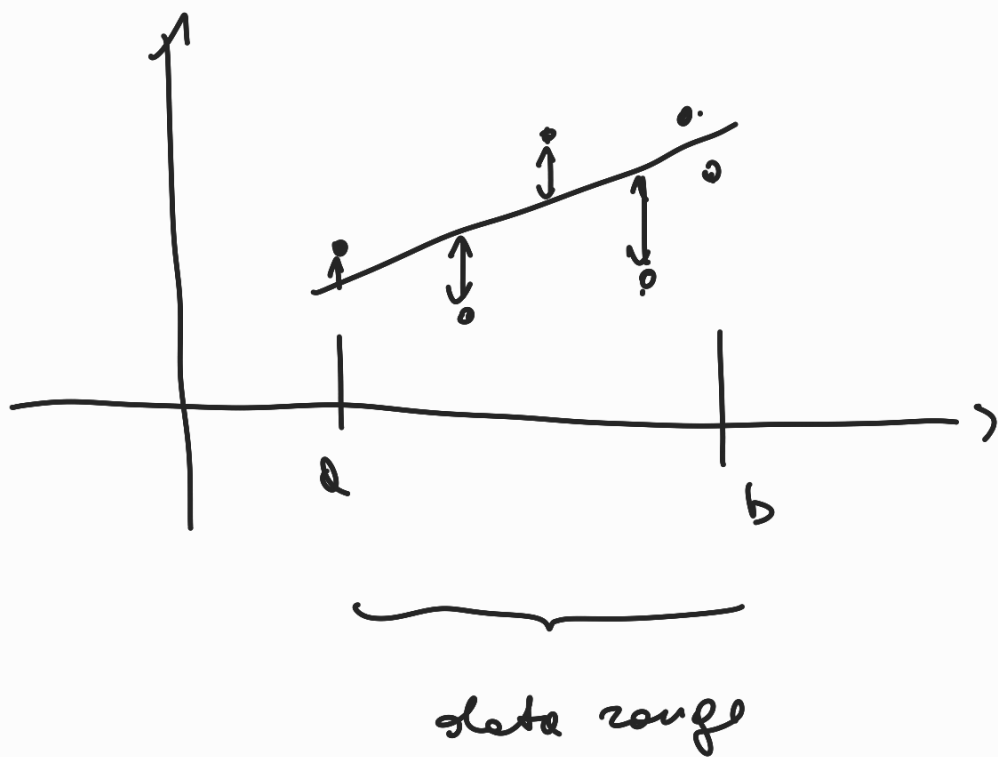
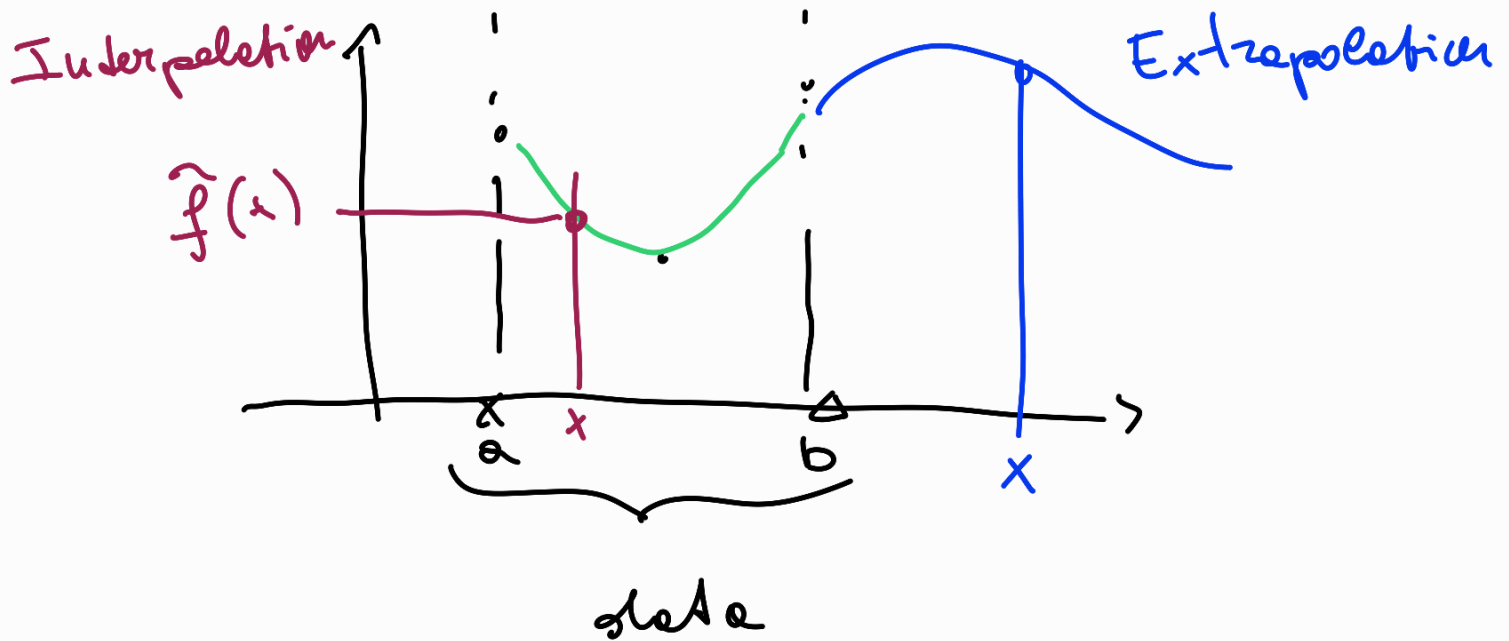
Least squares approximation

For polynomial interpolation the problem

$$\tilde{f}(x_j) = f(x_j) \quad j=0, \dots, n$$

Is well posed if the interpolation points are distinct and

$$\tilde{f} \in \mathbb{P}_n$$



Given data points $(x_i, y_i) \quad i = 0 \dots n$

we want to find $\tilde{f} \in \mathcal{P}_m$ s.t.

$$\sum_{i=0}^n [y_i - \tilde{f}(x_i)]^2 \leq \sum_{i=0}^n [y_i - p_m(x_i)]^2 \quad \forall p_m \in \mathcal{P}_m$$

Least squares approximation

$$\tilde{f}(x_j) \neq f(x_j) = y_j$$

$\tilde{f}$ minimizes the square sum error

$$\sum_{i=0}^n [p(x_i) - y_i]^2$$

This problem also has solution if $m < n$

Linear regression - regression line

$$\tilde{f}(x) = a_0 + a_1 x$$

$$\phi(a_0, a_1) = \sum_{i=0}^n (y_i - (a_0 + a_1 x_i))^2$$

the regression line $\tilde{f}$ for data set (x_i, y_i)

is defined

$$(a_0, a_1): \phi(b_0, b_1) = \min_{b_0, b_1} \phi(b_0, b_1)$$

$$a_0 = \frac{1}{D} \left[\sum_{i=0}^n y_i \sum_{j=0}^n x_j^2 - \sum_{j=0}^n x_j \sum_{i=0}^n x_i y_i \right]$$

$$a_1 = \frac{1}{D} \left[(n+1) \sum_{i=0}^n x_i y_i - \sum_{j=0}^n x_j \sum_{i=0}^n y_i \right]$$

$$D = (n+1) \sum_{i=0}^n x_i^2 - \left(\sum_{i=0}^n x_i \right)^2$$